

EdTech Island Platform

Comprehensive Technical Architecture, Database DDL, Multi-Modal Learning Engines, Gemini AI RAG, Realtime WebSockets & Evidence Analytics System

PRIMARY STACK

Node.js • React 18 • Three.js • Supabase

CLOUD OBJECT STORAGE

Cloudflare R2 (S3 API) + Public CDN

DATABASE & VECTOR

PostgreSQL 15 • pgvector (HNSW)

AI & REALTIME ENGINE

Google Gemini API • Native WebSockets (ws)

1. Executive Summary & Architectural Philosophy

EdTech Island is an immersive, multi-tenant K-12 STEM learning ecosystem architected to deliver CBSE-aligned science and mathematics education through spatial 3D exploration, virtual physics laboratories, and adaptive cognitive assessments.

The Core Axiom: *Raw Learner Action → Activity Telemetry → Curriculum/LO Evidence → Performance Metrics → Analytics Signals → Dashboard Interpretation & Gemini AI Remediation.*

1.1 The Ponytail Engineering Discipline

- **Zero-Waste Runtime:** Minimize unnecessary runtime bloat by leveraging native Web APIs (WebSockets, BroadcastChannel, Web Workers, CSS Glassmorphism).
- **The Client is Untrusted:** All critical identities, enrollments, authorization boundaries, and mastery metrics are computed on the server side via Edge Functions and database triggers.
- **Immutable Telemetry & Recomputability:** Raw event streams are append-only. When algorithmic weights evolve (e.g., `MASTERY_V1` to `MASTERY_V2`), historical records can be reprocessed without data loss.
- **O(1) Precomputed Dashboards:** Student and faculty dashboards query pre-aggregated read models (`dimension_profiles`, `lo_performance_snapshots`), eliminating expensive multi-table scans on live analytics views.



2. Complete Technology Stack Matrix

TIER / DOMAIN	TECHNOLOGIES USED	ARCHITECTURAL FUNCTION
Frontend Core	HTML5, CSS3, Vanilla ES6+ JS, React 18, Vite, Tailwind CSS	Instant loading in Study Island; reactive state machines in Teacher, Student, and Admin portals.
3D Graphics & Canvas	WebGL, Three.js, OrbitControls, GLTFLoader, Canvas 2D	Spatial simulations (e.g. shadow cone formation, raytracing optics, solar system orbits).
Backend & Middleware	Node.js <code>http</code> module, <code>ws</code> WebSocket library	Local & Edge server router, R2 upload proxy, JWT verification, real-time presence broadcasting.
Relational Database	Supabase (PostgreSQL 15), Row Level Security (RLS)	Primary relational storage for profiles, enrollments, curriculum, and telemetry evidence.
Vector Database	<code>pgvector</code> PostgreSQL extension (HNSW index)	768-dimensional semantic lesson embeddings for AI-driven doubt solving and remediation.
Object Storage & CDN	Cloudflare R2 (S3-Compatible API) + Public CDN	Zero-egress hosting for 3D <code>.gltf</code> assets, HTML5 lab packages, avatars, and streaming media.
Artificial Intelligence	Google Gemini API (<code>gemini-1.5-flash</code> / <code>2.0-flash</code>)	Contextual doubt resolution, automatic Bloom-level quiz generation, and diagnostic alert generation.
Production Hosting	Netlify Edge CDN / Local Node.js runtime	Hybrid deployment capability supporting cloud infrastructure and offline classroom setups.

3. Repository Directory Structure & Bundle

Blueprint

```
Platform CODE/
├─ server.js                # Central Node.js Server (Router, R2 proxy, WebSocket Presence)
├─ supabase-config.js       # Global Supabase client configuration & credentials
├─ package.json             # Root orchestrator with multi-portal build scripts
├─ index.html               # Public Marketing Landing Page
├─ login.html               # Universal SSO / Role-based Authentication Entrypoint
├─ superadmin-login.html    # Specialized SuperAdmin Gateway
├─ curriculum-migration.sql # PostgreSQL schema DDL & migrations
├─ .env                     # Environment secrets (Supabase, R2, Gemini, Port)

├─ study-island/           # ★ CORE IMMERSIVE STUDENT LEARNING ENGINE
│   ├── index.html          # Main Study Island HUD & Navigation Shell
│   ├── script.js           # Master Controller (3470+ lines: Modalities, Canvas, AI Chat)
│   ├── styles.css          # Cyber-Island Glassmorphic Theme & Responsive Layout
│   ├── Chapter_experience_L_S.html # 3D Three.js Interactive Visualizer (Light & Shadows)
│   ├── Shadow_Lab.html     # Interactive Physics Virtual Lab (Variable Manipulation)
│   ├── quiz.html           # Adaptive Quiz Engine & High-Stakes Challenge Mode
│   ├── perf_optimizer.js   # WebGL context lifecycle & memory management
│   └─ sw.js                # Service Worker for asset caching & offline resilience

├─ Teacher Portal/         # 🧑 TEACHER SINGLE PAGE APP (React 18 + Vite)
│   └─ src/
│       ├── main.jsx / App.jsx # Root React Component & Routing State Machine
│       ├── ThemeContext.jsx   # Dynamic Theme Switcher (Dark, Light, Cyber, Sunset)
│       └─ views/              # Core Feature Views:
│           ├── Dashboard.jsx  # Class KPIs, quick actions, schedule overview
│           ├── Classes.jsx    # Student rosters, attendance tracking, report cards
│           ├── SmartboardTeaching.jsx # Fullscreen classroom presentation projector view
│           ├── QuestionBank.jsx # Question Bank manager with Bloom & difficulty tags
│           ├── Analytics.jsx  # Heatmaps, LO mastery breakdown, divergence flags
│           ├── LiveClass.jsx  # Realtime classroom sessions with chat & polling
│           ├── TimeTable.jsx  # Interactive drag-and-drop weekly timetable
│           ├── Inbox.jsx      # Direct student & parent communication
│           └─ ToDoList.jsx    # Faculty task prioritization board

├─ CODE Message test/
│   └─ Student portal/     # 🎓 STUDENT DASHBOARD (React 18 + Vite)
│       └─ src/views/
│           ├── Dashboard.jsx  # Student summary, streak tracking, recent badges
│           ├── Courses.jsx    # Subject enrollment, curriculum syllabus navigator
│           ├── Progress.jsx   # 7-Dimension radar analytics, LO mastery metrics
│           ├── LiveClass.jsx  # Scheduled digital lectures & virtual classroom
│           ├── Mentors.jsx    # Faculty directory & office-hour requests
│           ├── Chats.jsx      # Realtime classmate messaging & study groups
│           └─ Timetable.jsx   # Class schedule calendar

│   └─ Admin Portal/       # 🏢 SCHOOL ADMIN ERP (React 18 + Vite)
│       └─ src/views/
│           ├── Dashboard.jsx  # School-wide attendance, fee alerts, active classes
│           ├── Classes.jsx    # Section allocation, teacher assignments
│           ├── Teachers.jsx   # Faculty recruitment, workload & payroll management
│           ├── Communications.jsx # School-wide SMS/Email/App announcements
│           ├── Events.jsx     # Academic calendar & examination scheduler
│           └─ Analytics.jsx   # Aggregate institution-level academic KPIs

├─ superadmin/             # 🔑 MULTI-TENANT SUPERADMIN PANEL
│   └─ index.html           # Single-file master console for school provisioning

├─ admin/ & teacher/ & student/ # Compiled production distributions (`dist/`)
└─ assets/ & shared/        # Shared static media, logos, audio, avatar presets
```

4. Deep Dive: The 6 Platform Portals

1. Study Island (/study-island/)

Primary student immersion HUD. Houses the 5 core modalities, subject carousel, 3D particle canvas, dynamic course builder, and floating Gemini AI Doubt Solver.

VANILLA ES6+

THREE.JS

2. Teacher Portal (/teacher/)

Faculty cockpit. Includes Smartboard projector mode, AI lesson/quiz generator, Question Bank manager, class analytics heatmap, and drag-and-drop timetable planner.

REACT 18

VITE SPA

3. Student Portal (/student/)

Academic summary dashboard. Displays 7-dimension cognitive radar charts, peer study group chats, mentor office hours, and live lecture joining.

REACT 18

RADAR CHARTS

4. Admin Portal (/admin/)

School ERP. Manages grade/section allocations, teacher-subject mapping, institutional attendance, academic event scheduling, and broadcast announcements.

REACT 18

ERP ENGINE

5. SuperAdmin Portal (/superadmin/)

Multi-tenant command center. Provisions new schools, monitors system-wide audit logs, and pushes canonical curriculum updates to all instances.

MULTI-TENANT

AUDIT LOGS

6. Marketing & SSO Gateways (/login.html)

Role-based single sign-on entrypoint. Validates credentials with Supabase Auth and redirects students, teachers, admins, and superadmins to their assigned portal.

AUTH ROUTER

JWT GUARD

EdTech Island — Canonical Architecture & Developer Specification

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5. The 5 Interactive Learning Modalities

5.1 Modality 1: 3D Interactive Chapter Experience (`Chapter_experience_L_S.html`)

An interactive WebGL simulation built with Three.js. For the *Light & Shadows* chapter, the scene renders directional raytraced light sources, an opaque spherical occluder, and a projection plane. Students manipulate spatial coordinates (x , y , z) in real time to observe the geometrical cone of the umbra (total shadow) and penumbra (partial shadow).

5.2 Modality 2: Virtual Experiments Lab (`Shadow_Lab.html`)

A physics laboratory simulator structured as a hypothesis-testing pipeline. Telemetry tracks:

- **Procedure Accuracy:** Adherence to proper sequence of experimental steps.
- **Variable Manipulation:** Changing distance d , light intensity I , or object diameter D .
- **Prediction Accuracy:** Formulating a scientific hypothesis prior to executing physical changes.

5.3 Modality 3: Adaptive Quiz & Challenge Mode (`quiz.html`)

Provides two distinct assessment environments powered by CBSE-tagged questions:

- **Zen Mode:** Untimed assessment focusing on deep comprehension. Integrates multi-level progressive hints and detailed explanations.
- **Challenge Mode:** High-pressure cognitive retrieval under strict timers. Evaluates response fluency, consecutive streak resilience, and stability under time constraints.

5.4 Modality 4: Curated Visual Stories & Video Hub

Video player tracking watch duration, pause frequency, speed adjustments, and segment rewinding. Rewind hotspots are aggregated to pinpoint confusing concepts.

5.5 Modality 5: Mixed Reality / WebXR Modalities

WebXR-compliant hooks enabling students to project 3D models into their physical environment using mobile AR or XR headsets.

6. Database Schema & Data Model (PostgreSQL 15)

6.1 Identity, Academic Context & Curriculum Backbone DDL

```
-- 1. PROFILES & ROLES
CREATE TABLE profiles (
  id          UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,
  email       TEXT UNIQUE NOT NULL,
  name        TEXT NOT NULL,
  role        TEXT NOT NULL CHECK (role IN ('student', 'teacher', 'admin', 'superadmin')),
  avatar_url  TEXT,
  department  TEXT,
  is_archived BOOLEAN DEFAULT false,
  timetable   JSONB,
  created_at  TIMESTAMPTZ DEFAULT now()
);

-- 2. ACADEMIC YEARS & ENROLLMENTS (Multi-Year Context)
CREATE TABLE academic_years (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name        TEXT NOT NULL UNIQUE, -- '2026-27'
  start_date  DATE NOT NULL,
  end_date    DATE NOT NULL,
  is_current  BOOLEAN DEFAULT false,
  created_at  TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE enrollments (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  student_id  UUID NOT NULL REFERENCES profiles(id) ON DELETE CASCADE,
  class_id    UUID NOT NULL REFERENCES classes(id) ON DELETE CASCADE,
  academic_year_id UUID NOT NULL REFERENCES academic_years(id) ON DELETE CASCADE,
  enrolled_at TIMESTAMPTZ DEFAULT now(),
  status      TEXT DEFAULT 'active' CHECK (status IN ('active', 'transferred', 'graduated')),
  UNIQUE (student_id, class_id, academic_year_id)
);

-- 3. CURRICULUM HIERARCHY & GRANULAR LEARNING OBJECTIVES
CREATE TABLE curriculum_sections (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  chapter_id  UUID REFERENCES course_chapters(id) ON DELETE CASCADE,
  section_order SMALLINT NOT NULL,
  code        TEXT NOT NULL, -- 'S01', 'S02'
  title       TEXT NOT NULL,
  created_at  TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE learning_objectives (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  chapter_id  UUID REFERENCES course_chapters(id),
  section_id  UUID REFERENCES curriculum_sections(id) ON DELETE CASCADE,
  code        TEXT NOT NULL UNIQUE, -- 'SCI6-CH03-S02-LO01'
  title       TEXT NOT NULL,
  description  TEXT,
  bloom_level TEXT NOT NULL CHECK (bloom_level IN
('remember', 'understand', 'apply', 'analyze', 'evaluate', 'create')),
  is_active   BOOLEAN DEFAULT true,
  created_at  TIMESTAMPTZ DEFAULT now()
);

-- 4. QUESTION BANK WITH BLOOM TAXONOMY & WEIGHTED LO MAPPINGS
CREATE TABLE question_bank (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  chapter_id  UUID REFERENCES course_chapters(id),
```

```

section_id    UUID REFERENCES curriculum_sections(id),
lo_id        UUID REFERENCES learning_objectives(id),
type         TEXT NOT NULL DEFAULT 'mcq',
text         TEXT NOT NULL,
options      JSONB NOT NULL,          -- Array of { id, text, is_correct, explanation }
difficulty   TEXT NOT NULL CHECK (difficulty IN ('easy', 'medium', 'hard')),
bloom_level  TEXT NOT NULL,
points       INTEGER DEFAULT 1,
version      INTEGER DEFAULT 1,
is_active    BOOLEAN DEFAULT true,
created_at   TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE question_learning_objectives (
question_id   UUID REFERENCES question_bank(id) ON DELETE CASCADE,
lo_id        UUID REFERENCES learning_objectives(id) ON DELETE CASCADE,
weight       NUMERIC(3,2) DEFAULT 1.0,
PRIMARY KEY (question_id, lo_id)
);

```


6.2 Universal Analytics Engine Schema DDL

```
-- 5. ACTIVITY ATTEMPTS (Server-Generated Attempt Session)
CREATE TABLE activity_attempts (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  student_id        UUID NOT NULL REFERENCES profiles(id),
  enrollment_id      UUID REFERENCES enrollments(id),
  activity_type      TEXT NOT NULL CHECK (activity_type IN
('platform_quiz','challenge_quiz','chapter_experience','experiment','video','teacher_assessment')),
  activity_id        UUID NOT NULL,
  activity_version   INTEGER NOT NULL DEFAULT 1,
  chapter_id         UUID REFERENCES course_chapters(id),
  attempt_number     INTEGER NOT NULL DEFAULT 1,
  started_at         TIMESTAMPTZ NOT NULL,
  completed_at       TIMESTAMPTZ,
  status             TEXT DEFAULT 'started' CHECK (status IN ('started','completed','abandoned','paused')),
  raw_result         JSONB,
  processed          BOOLEAN DEFAULT false,
  created_at         TIMESTAMPTZ DEFAULT now(),
  UNIQUE (student_id, activity_id, activity_version, attempt_number)
);

-- 6. IMMUTABLE RAW TELEMETRY EVENTS
CREATE TABLE analytics_events (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  attempt_id         UUID REFERENCES activity_attempts(id),
  student_id         UUID NOT NULL,
  event_type         TEXT NOT NULL,
  event_sequence     INTEGER NOT NULL,
  payload            JSONB,
  event_timestamp    TIMESTAMPTZ DEFAULT now(),
  created_at         TIMESTAMPTZ DEFAULT now()
);
CREATE UNIQUE INDEX idx_ae_idempotency ON analytics_events(attempt_id, event_sequence) WHERE attempt_id IS NOT NULL;

-- 7. NORMALIZED MEASUREMENTS (Layer 2)
CREATE TABLE activity_measurements (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  attempt_id         UUID REFERENCES activity_attempts(id) ON DELETE CASCADE,
  student_id         UUID NOT NULL,
  lo_id             UUID REFERENCES learning_objectives(id),
  measurement_type    TEXT NOT NULL, -- 'accuracy', 'response_time_ms', 'procedure_accuracy', etc.
  measurement_value   NUMERIC(8,4) NOT NULL,
  unit               TEXT,           -- 'pct', 'ms', 'count', 'bool'
  difficulty         TEXT,
  bloom_level        TEXT,
  hint_used          BOOLEAN DEFAULT false,
  retry_count        SMALLINT DEFAULT 0,
  independence_level TEXT CHECK (independence_level IN ('independent','assisted','retry_corrected','failed')),
  rule_version        TEXT DEFAULT 'v1.0',
  created_at         TIMESTAMPTZ DEFAULT now()
);

-- 8. LEARNING EVIDENCE (Layer 3)
CREATE TABLE learning_evidence (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  student_id         UUID NOT NULL,
  lo_id             UUID REFERENCES learning_objectives(id),
  dimension          TEXT NOT NULL CHECK (dimension IN
('mastery','fluency','application','exploration','practical_skill','engagement','formal_achievement')),
  evidence_value      NUMERIC(5,2) NOT NULL, -- 0.00 to 100.00
  confidence          TEXT NOT NULL CHECK (confidence IN ('low','medium','high')),
  independence_level TEXT,
  source_type         TEXT NOT NULL,
  source_attempt_id   UUID REFERENCES activity_attempts(id),
  observed_at        TIMESTAMPTZ DEFAULT now(),
);
```

```

    rule_version          TEXT DEFAULT 'v1.0',
    created_at            TIMESTAMPTZ DEFAULT now()
);

-- 9. PRECOMPUTED READ MODEL: DIMENSION PROFILES
CREATE TABLE dimension_profiles (
    id                    UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    student_id           UUID NOT NULL,
    lo_id                UUID REFERENCES learning_objectives(id),
    dimension             TEXT NOT NULL,
    current_value         NUMERIC(5,2),
    confidence            TEXT,
    recent_value          NUMERIC(5,2),      -- 7-Day Rolling Window
    monthly_value         NUMERIC(5,2),      -- 30-Day Window
    historical_value       NUMERIC(5,2),      -- All-Time Weighted Average
    trend                 TEXT CHECK (trend IN
('INSUFFICIENT_EVIDENCE', 'STABLE', 'SLIGHT_IMPROVEMENT', 'MEANINGFUL_IMPROVEMENT', 'MAJOR_IMPROVEMENT', 'SLIGHT_DE
CLINE', 'MEANINGFUL_DECLINE', 'MAJOR_DECLINE')),
    active_signals        TEXT[],
    updated_at            TIMESTAMPTZ DEFAULT now(),
    UNIQUE (student_id, lo_id, dimension)
);

-- 10. AI-AUGMENTED VECTOR EMBEDDINGS (pgvector)
CREATE TABLE lesson_embeddings (
    id                    UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    lesson_id            TEXT NOT NULL,
    lesson_title          TEXT NOT NULL,
    content_text          TEXT NOT NULL,
    embedding             vector(768),      -- Google text-embedding-004
    created_at            TIMESTAMPTZ DEFAULT now()
);
CREATE INDEX idx_lesson_embed_hnsw ON lesson_embeddings USING hnsw (embedding vector_cosine_ops);

```

7. Backend Server Architecture & WebSocket Presence (`server.js`)

The Node.js server (`server.js`) acts as the local edge runtime, serving static bundles, proxying authenticated Cloudflare R2 uploads, and maintaining a high-performance native WebSocket presence engine.

7.1 Native WebSocket Presence Engine

```
const { WebSocketServer } = require('ws');
const wss = new WebSocketServer({ server });
const activePresenceMap = new Map(); // ws socket -> email

const broadcastPresenceList = () => {
  const onlineEmails = Array.from(new Set(Array.from(activePresenceMap.values()).filter(Boolean)));
  const payload = JSON.stringify({ type: 'presence_sync', emails: onlineEmails });
  for (const client of wss.clients) {
    if (client.readyState === 1) client.send(payload);
  }
};

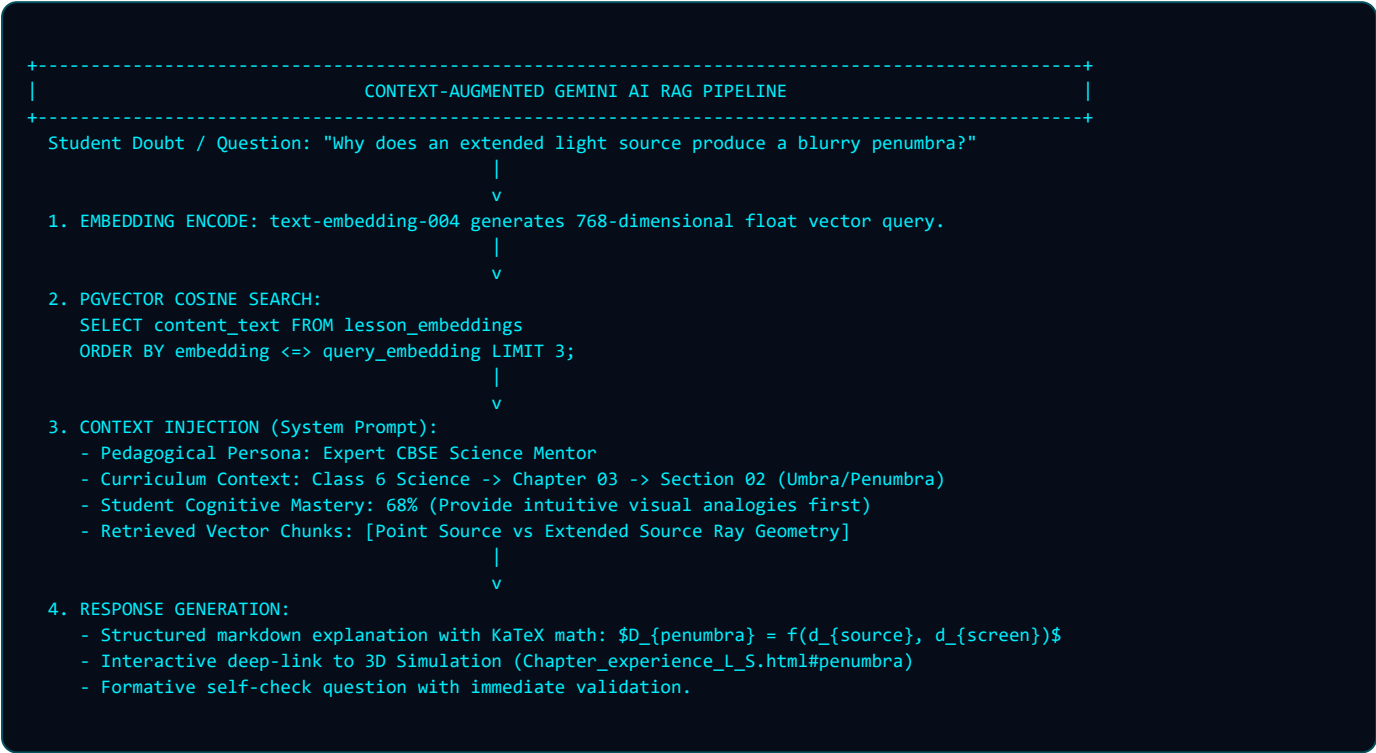
wss.on('connection', (ws) => {
  ws.on('message', (msg) => {
    try {
      const data = JSON.parse(msg.toString());
      if (data.type === 'identify') {
        if (data.email) activePresenceMap.set(ws, data.email.toLowerCase().trim());
        else activePresenceMap.delete(ws);
        broadcastPresenceList();
      } else if (data.type === 'timetable_update') {
        const payload = JSON.stringify({ type: 'timetable_update', payload: data.payload });
        for (const client of wss.clients) {
          if (client.readyState === 1) client.send(payload);
        }
      }
    } catch (e) {}
  });
  ws.on('close', () => { activePresenceMap.delete(ws); broadcastPresenceList(); });
  ws.on('error', () => { activePresenceMap.delete(ws); broadcastPresenceList(); });
});
```

7.2 Cloudflare R2 S3 Upload Endpoint (`/api/upload-r2`)

- **JWT Verification:** Validates Bearer token via Supabase Auth API (`/auth/v1/user`).
- **Deterministic Key Resolution:** `courses/{class}/{subject}/{chapter}/{modality}/{file}` or `avatars/{cleanEmail}_{timestamp}_{file}`.
- **Payload Ceiling:** Enforces a strict 50MB binary ceiling with automatic MIME detection.

8. Artificial Intelligence Engine (Google Gemini + pgvector)

Gemini AI is situated strictly *above the evidence layer*. It synthesizes structured telemetry outputs (`dimension_profiles` , `analytics_signals`) and textbook vector embeddings rather than directly consuming raw clicks.



8.1 Teacher AI Co-Pilot & Diagnostic Alert Synthesis

When an `APPLICATION_GAP` signal is detected (Mastery > 70% but Experiment Application < 50%), Gemini generates an actionable instructional note for the teacher:

Sample Teacher Diagnostic Note:

"Priya demonstrates high conceptual knowledge in ray optics (84% Mastery, HIGH confidence), but experienced difficulty during the Shadow Lab experiment when predicting shadow length changes at acute angles (32% Application). Recommended Action: Have Priya re-attempt Step 3 of the Virtual Lab, focusing on the inverse relationship between distance and umbra magnitude."

9. The Universal Analytics Engine

9.1 The 3-Layer Evidence Pipeline

PIPELINE LAYER	PRIMARY DATABASE TABLE	TRANSFORMATION LOGIC & OPERATIONAL LATENCY
Layer 1: Raw Telemetry	analytics_events	Immutable record of user actions. Fire-and-forget async POST (< 50ms).
Layer 2: Measurements	activity_measurements	Standardized metrics per LO. Resolves independence level (< 500ms via DB trigger).
Layer 3: Learning Evidence	learning_evidence	Weighted synthesis across 7 dimensions and 5 longitudinal windows (≤ 5 min via cron).

9.2 The 7 Evaluated Dimensions

 **1. Mastery**
Core conceptual understanding. Calculated via weighted correctness across difficulty and Bloom taxonomy levels.

 **2. Fluency**
Retrieval speed, consecutive correct streaks, and resilience under countdown timers in Challenge Mode.

 **3. Application**
Performance on `Apply` / `Analyze` Bloom questions combined with experiment prediction accuracy.

 **4. Practical Skill**
Scientific laboratory execution: procedure sequencing, variable manipulation, and outcome verification.

 **5. Exploration**
Depth of engagement in 3D Three.js scenes: required milestones reached vs optional interactive objects examined.

 **6. Engagement**
Longitudinal study regularity, session frequency, video completion rate, and re-engagement streaks.

9.3 Mathematical Calculation Models (Versioned Heuristics)

A. Independence Classification (`INDEPENDENCE_V1`)

```
IF hint_used = false AND retry_count = 0 AND attempt_number = 1:
  independence_level = 'independent'      weight_multiplier = 1.00
ELSE IF hint_used = true AND correct = true:
  independence_level = 'assisted'          weight_multiplier = 0.60
ELSE IF retry_count > 0 AND correct = true:
  independence_level = 'retry_corrected'   weight_multiplier = 0.75
ELSE IF correct = false:
  independence_level = 'failed'            weight_multiplier = 0.00
```

B. Mastery Evidence Score Formula (`MASTERY_V1`)

```
Mastery_Score = [ SUM( Correct * Difficulty_Weight * Bloom_Weight * Independence_Multiplier ) /  
                  SUM( Difficulty_Weight * Bloom_Weight ) ] * 100  
  
Difficulty_Weights: Easy = 0.60 | Medium = 1.00 | Hard = 1.40  
Bloom_Weights:      Remember = 0.50 | Understand = 0.70 | Apply = 1.00 | Analyze = 1.20 | Evaluate = 1.40 |  
Create = 1.50
```

C. Fluency Evidence Score Formula (`FLUENCY_V1`)

```
Fluency_Score = ( Speed_Score * 0.35 ) + ( Independence_Score * 0.40 ) + ( Pressure_Score * 0.25 )  
Where: Speed_Score = CLAMP( 1 - ( avg_response_ms / 30000 ), 0, 1 ) * 100
```

10. Developer Quickstart, Build Scripts & Deployment

10.1 Environment Variables (`.env`)

```
PORT=3000
VITE_SUPABASE_URL=https://qmyrxvtbzlbnvzxypnus.supabase.co
VITE_SUPABASE_ANON_KEY=eyJhbGciOi...
CLOUDFLARE_R2_ACCESS_KEY_ID=your_access_key
CLOUDFLARE_R2_SECRET_ACCESS_KEY=your_secret_key
CLOUDFLARE_R2_BUCKET_NAME=edtechplatform
CLOUDFLARE_R2_PUBLIC_URL=https://pub-670b98370fe642a2be08ee37cbfd385f.r2.dev
CLOUDFLARE_R2_ENDPOINT=https://[account_id].r2.cloudflarestorage.com
```

10.2 Build Scripts

```
# Install Root Dependencies
npm install

# Compile all React SPAs (Teacher, Student, Admin, Study Island)
npm run build:all

# Start Local Hybrid Node.js & WebSocket Server
npm start

# -> Access at http://localhost:3000/
```

10.3 Core Portal Endpoints Reference

PORTAL ROUTE	SOURCE DIRECTORY	INTENDED USER BASE
<code>http://localhost:3000/index.html</code>	<code>/index.html</code>	Public Marketing & Parents
<code>http://localhost:3000/login.html</code>	<code>/login.html</code>	Universal Role-Based SSO Gateway
<code>http://localhost:3000/study-island/</code>	<code>/study-island/</code>	Active Students (3D & Labs)
<code>http://localhost:3000/teacher/</code>	<code>/Teacher Portal/</code>	Teachers (Smartboard & AI Tools)
<code>http://localhost:3000/student/</code>	<code>/CODE Message test/Student portal/</code>	Students (Progress & Community)
<code>http://localhost:3000/admin/</code>	<code>/CODE Message test/Admin Portal/</code>	School Principals & Staff
<code>http://localhost:3000/superadmin/</code>	<code>/superadmin/index.html</code>	Multi-Tenant Platform Owners

Verification Note: This document represents the canonical system specification for EdTech Island v3.0. All subsequent code modifications and database schema migrations must preserve compatibility with the DDL, API contracts, and analytics evidence formulas defined herein.